

# Class I glucose-6-phosphate dehydrogenase deficiency

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Pipeline complete — awaiting your sign-off

## Repurposing hypothesis: BREXANOLONE → Class I glucose-6-phosphate dehydrogenase deficiency

**NOTE:** This candidate did NOT meet the STRONG\_MATCH threshold (composite  $0.4000 < 0.70$ ). It is included as the highest-ranked hypothesis for review; treat it accordingly.

**Repurposing-only pool:** the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max\_phase  $\geq 4$ ) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

## 1. Hypothesis summary

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BREXANOLONE is proposed as a repurposing candidate against **Class I glucose-6-phosphate dehydrogenase deficiency** via the target **G6PD**. It shows a ChEMBL median pChEMBL affinity of 4.18 at assay confidence 9/9, an Open Targets target-disease association of 0.851, and a Tanimoto similarity of 0.259 to PRASTERONE. Target network context (BioGRID, physical/genetic — not mechanism): G6PD, HSPB1, HARS, CCT4, FN1, BAG3, ALDH7A1, CAPN1. The resulting composite score is 0.4000.

*Chemist rationale:* Brexanolone is an approved drug with a median pChEMBL affinity of 4.18 against G6PD, measured at assay confidence score 9 out of 9. Its Tanimoto similarity (Morgan r2) to the nearest approved drug, prasterone, is 0.259, and G6PD has eight recorded BioGRID interactors: HSPB1, HARS, CCT4, FN1, BAG3, ALDH7A1, CAPN1, and G6PD itself.

## STAGE 1 PRIORITIZATION SCORES

- **tractability\_score:** 0.5898 (ChEMBL bioactivity + AlphaFold pLDDT – prior-trial-failure penalty)
- **unmet\_need\_score:** 1.0000 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

## 2. Evidence table

| Evidence                                           | Value                                             |
|----------------------------------------------------|---------------------------------------------------|
| ChEMBL median pChEMBL affinity                     | 4.18                                              |
| Assay confidence score (0-9)                       | 9                                                 |
| Open Targets association score                     | 0.851                                             |
| Tanimoto to nearest approved drug                  | 0.259 (PRASTERONE)                                |
| Approved / known drug                              | yes                                               |
| Mutation-specific approved indication (disclosure) | No specific mutation named in approved indication |
| Lipinski/Veber (MW, logP, HBD, HBA,                | 318.5, 4.60, 1, 2, 37.3, 1                        |

|                                                                                |                                                                                                                                        |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| TPSA, rotB)                                                                    |                                                                                                                                        |
| Lipinski violations<br>/ Veber pass                                            | 0 / yes                                                                                                                                |
| AFDB apo structure<br>mean pLDDT (free<br>protein, no ligand)                  | 94.4                                                                                                                                   |
| Boltz structure<br>confidence (0-1)                                            | 0.770                                                                                                                                  |
| Boltz binding-pose<br>confidence (0-1)                                         | 0.388                                                                                                                                  |
| Boltz predicted<br>affinity (relative<br>optimization score,<br>0-1, NOT a Kd) | 0.089                                                                                                                                  |
| Boltz predicted<br>structure (CIF)                                             | <a href="/api/structures/04939a92a52c935a420ad69f6fb40da9.cif">Download CIF (/api/structures/04939a92a52c935a420ad69f6fb40da9.cif)</a> |
| Boltz ADME –<br>lipophilicity (logD)                                           | 4.092                                                                                                                                  |
| Boltz ADME –<br>permeability                                                   | 1.202                                                                                                                                  |
| Boltz ADME –<br>solubility                                                     | high-risk                                                                                                                              |
| openFDA adverse-<br>event signal (FAERS)                                       | ACUTE LUNG INJURY (2),<br>BRADYCARDIA (2), DIZZINESS (2),<br>HYPOTENSION (2), MULTIPLE ORGAN<br>DYSFUNCTION SYNDROME (2)               |
| Prior trials for                                                               | 0                                                                                                                                      |

|                            |                     |
|----------------------------|---------------------|
| this exact<br>drug+disease |                     |
| Target discovery<br>method | genetic_association |

### 3. Full source citations

- PMIDs (0): none
- ChEMBL activity IDs (1): 10959852
- NCT numbers (0): none

### 4. Composite score breakdown

| Term                                | Weight | Component value | Contribution |
|-------------------------------------|--------|-----------------|--------------|
| Normalized pChEMBL affinity         | 0.30   | 0.169           | 0.0506       |
| Assay confidence (score / 9)        | 0.20   | 1.000           | 0.2000       |
| Normalized Open Targets association | 0.20   | 0.851           | 0.1702       |
| Normalized Tanimoto similarity      | 0.15   | 0.259           | 0.0388       |
| No prior failed trial (1/0)         | 0.15   | 1               | 0.1500       |
|                                     |        |                 |              |

|                                                         |   |   |         |
|---------------------------------------------------------|---|---|---------|
| Safety cap (ChEMBL safety signal, hard gate, max 0.400) | – | – | applied |
| Composite (weighted sum - penalty - cap)                |   |   | 0.4000  |

Weighted sum before penalty = 0.6096; penalty = 0.0000; reported composite\_score = 0.4000. Safety cap applied (capped at 0.400) — WITHDRAWN FOR SAFETY (ChEMBL structured data) — [https://www.ebi.ac.uk/chembl/compound\\_report\\_card/CHEMBL207538/](https://www.ebi.ac.uk/chembl/compound_report_card/CHEMBL207538/)

## 5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.388) or predicted affinity (0.089) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.
- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure\_confidence of 0.770 (complex pLDDT 0.823) and an AFDB apo mean pLDDT of 94.4; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.